

Aliaksei Ivanou | C++ Software Engineer

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Poland: National D work visa valid from 26 Sep 2026; employer-registered oświadczenie and signed umowa required

C++/Qt Software Engineer with 6 years of commercial C++ experience across systems, embedded Linux, and UI/HMI development, plus 7+ years in R&D/DSP. Strong hands-on experience with C++11-17, Qt 5.15.7, Qt Widgets/QSS, SIP phone UI implementation, VoIP/WebRTC workflows, and embedded Linux integration. Proven track record in embedded UI implementation, SIP/RTP telephony, cross-platform tooling, and performance-sensitive C++ development.

Tech Stack & Domains

- Primary Languages & Frameworks: C++11-17, C89, MATLAB, Java 8/11, Scala, Qt 5.15.7, Qt Widgets, Qt Designer .ui forms, QSS, Qt Model/View, QTest, Google Test, FLTK, Point Cloud Library.
- Interoperability & Scripting: C# (.NET 8), WinForms, Node.js, JavaScript.
- Databases & Storage: PostgreSQL, MySQL, SQLite, MS Access.
- Embedded, DevOps & Infrastructure: Linux, Yocto/BitBake Embedded Linux, NXP i.MX, AWS EC2, Azure VMs, Apache, SSH, Jenkins/GitLab CI, CI/CD pipelines.
- Networking, VoIP & Media: SIP, RTP/SDP, WebRTC, SNMP, TCP/IP, WebSockets, DHCP, TFTP, Linphone-SDK/liblinphone, mediasoup/libmediasoupclient, GStreamer, V4L2.
- Tools & Practices: Git, GitHub, BitBucket, qmake, CMake, GCC, gdb, Valgrind, Qt Creator, Qt Designer, plog, TDD, Waterfall, Proprietary Software.
- Embedded UI/HMI: Touch-oriented SIP phone UI, small-screen layouts, video-call/conference screens, custom Qt widgets, QSS styling, localization, touch-screen UX constraints.
- Domains: VoIP/SIP Telephony, Embedded SIP Phone UI/HMI, Embedded Video Conferencing, Hardware Interoperability, Aerospace & Satellite Systems, UAV Systems, Geoservices, Hardware Engineering, Industrial Software, Robotics, FinTech, Energy Sector ERP, Digital Library Systems.

Experience

C++ Software Engineer | [Innowise Group](#)

12/2024 - Present

Innowise is a global software development company, specializing in Big Data, AI, RPA, IoT, AR/VR, and Blockchain.

- Refactored SIP call handling toward a cleaner MVC/MVVM-style architecture by separating UI rendering from call state, user actions, media negotiation, and Linphone-SDK integration logic.
- Incrementally decomposed legacy Qt Widgets call-flow code into clearer domain, presentation, and SDK-integration boundaries while preserving existing UI behavior and QObject signal compatibility.
- Hardened video-call and conference UI behavior for edge cases such as held calls, second-line calls, remote hangups, stale overlays, and active video stream switching.
- Extended and customized Linphone-SDK (liblinphone) backend logic to implement SIP failover and backup-server connectivity for improved device reliability.
- Developed WebRTC video-conferencing UI and signaling flows: conference creation/join, incoming invite screens, participant roster, SIP/IP/phonebook add dialogs, status strip, control panel, video tile layout, and audio/video device controls.
- Configured and deployed mediasoup server on embedded Linux as a systemd-managed service, including runtime configuration and deployment automation.
- Autoprovisioning tooling: migrated a WinForms-based configuration utility from .NET Framework 4.8 to .NET 8, unified project file formats across the device ecosystem, and implemented provisioning workflows using SNMP, DHCP, TFTP, and SQLite.
- Collaborated on Yocto/BitBake customization, i.MX embedded Linux target integration, and device-side service configuration for hardware-software integration.
- Added and maintained QTest-based unit/integration tests for SIP call flows, media-option parsing, direct-IP/SIP validation, networking validators, and Linphone-dependent scenarios, including offscreen test execution on the embedded target.
- Supported resource planning and mentoring as the engineering sub-group grew to 6 people: helped with task allocation, onboarding, technical guidance, and candidate interviews while remaining primarily hands-on.

Tech Stack & Domains: C++17, Qt 5.15.7, Qt Widgets, Qt Designer .ui forms, QSS, Qt Model/View, QTest, Qt WebSockets, Linphone-SDK, liblinphone, Node.js, mediasoup, libmediasoupclient, WebRTC, GStreamer, V4L2, SIP/RTP/SDP Interoperability, Yocto, BitBake, NXP i.MX, C# (.NET 8), WinForms, SQLite, SNMP, DHCP, TFTP, GitLab CI, gdb | Embedded SIP Phone UI/HMI, VoIP/SIP Telephony, Embedded Video Conferencing, Hardware Interoperability.

(Promoted from Trainee to Mid-level/Senior roles within the 4-year tenure)

EPAM Systems (NYSE: EPAM) is one of the global leaders in both digital engineering and IT consulting, recognized for innovation, cloud, and AI services.

- **FinTech:** Engineered a high-performance Financial Analytics System (C++17) using FLTK and SQLite; established full CI/CD lifecycle with Google Test and CMake.
- Developed C++ modules for two consecutive enterprise projects: an Energy ERP System (C++17) and a legacy Digital Library Platform (C89).
- Managed remote debugging and deployment for C/C++ codebases on AWS EC2 and Azure VMs using SSH and gdb for core dump analysis.
- **Digital Library:** Maintained legacy C89 modules and resolved complex integration defects between the C core and JVM-based services (Scala/Java).
- **Energy ERP:** Optimized PostgreSQL schema and SQL queries, reducing search operation response times from **1.5s to 300ms**.
- Contributed as a Full-stack engineer during delivery phases, maintaining JavaScript modules for data flow between the C++ backend and Web UI.

Tech Stack & Domains: C (C89), C++ (14/17), Java (8/11), Scala, JavaScript, Amazon Web Services (AWS), Microsoft Azure, Apache, Linux, gdb, SSH, git, BitBucket, Jenkins, GCC, Valgrind, CMake, TDD, CI/CD pipelines, PostgreSQL, MySQL, TCP/IP, WebSockets | Energy Sector ERP System, Digital Library Platform.

C++ Software Engineer | [RIFTEK](#)

03/2020 - 05/2020

RIFTEK develops optoelectronic instruments for measuring geometrical quantities, including laser sensors, 2D/3D scanners, and machine vision systems, serving customers in over 70 countries.

- **R&D:** Developed a 3D reconstruction prototype using Point Cloud Library (PCL) and Qt, facilitating the transition from MATLAB-based research to C++ production code.
- Implemented C++ applications with the Point Cloud Library on Qt.

Tech Stack & Domains: C++17, git, GitHub, Qt, Point Cloud Library, TDD, MySQL | Industrial Software, Robotics.

MATLAB Algorithm Developer | [PELENG](#)

08/2012 - 01/2020

PELENG is a developer and manufacturer of high-tech optical and optoelectronic systems for space, industrial, and surveillance applications. The company specializes in thermal imaging, precision optics, and advanced vision systems, emphasizing innovation and collaboration with public and private sectors.

- Developed Earth Remote Sensing algorithms; implemented feature-based image stitching using keypoint matching and defect removal to optimize post-processing workflows.
- Automated resolution assessment for flight tests, replacing manual analysis with MATLAB-based processing modules.
- Implemented a custom lossy image compression algorithm (MATLAB) for satellite transmission, achieving a **reliable 3-4x compression ratio** while carefully balancing integrity preservation and computational constraints.
- Developed software in MATLAB for image type prediction for image quantization, improving image quality metrics by ~30% and enhancing Earth observation accuracy by an estimated 15-20%.
- Developed image processing modules for UAVs (MATLAB), including debayering, stabilization, and sensor fusion.

Tech Stack & Domains: MATLAB, MS Access, git, GitHub, Proprietary Software, TDD, Waterfall | Geoservices, Hardware engineering, Industrial software.

Education

- Bachelor's Degree in Radiophysics. Satellite information systems and technologies: [Belarusian State University, Faculty of Radiophysics and Computer Technologies](#) | 2007 - 2012

Languages

- **English:** Professional Working Proficiency (B2)
- **Polish:** Conversational proficiency (B1)
- **Swedish:** Elementary proficiency (A2)
- **Russian:** Native proficiency (C2)
- **Belarusian:** Native proficiency (C2)